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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester M.Tech (CSE) (CBS) (CBCS) Degree Examinations

December 2024- January 2025

23CSE114/22CBS133 – DESIGN THINKING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	How does identifying the right problem influence the effectiveness of a design thinking project?	6	L*2	1	1
b)	What are the three stages of journey mapping, and how does understanding these stages help design thinking practitioners?	6	L1	1	1
c)	What are the Empathy and Prototype modes in design thinking, and why are they important? Explain their significance in the design process.	8	L2	1	1
2. a)	Explain the purpose of a mission statement in the context of a product development project for the cordless screwdriver project.	6	L2	2	1
b)	Discuss the advantages and disadvantages of using interviews versus focus groups for collecting customer data. When might one method be preferred over the other?	6	L1	2	1
c)	Describe the step-by-step procedure for organizing customer needs into a hierarchical list.	8	L2	2	2
3. a)	What is design thinking? Explain four pillars of design thinking in detail.	10	L1	1	2
b)	How effectively solid waste management can be done? Explain using the five phases of the design thinking process.	10	L2	2	1
Unit – II					
4. a)	How does QFD facilitate the alignment of product development with customer needs? Include examples to illustrate your points.	10	L2	3	1
b)	What is needs-matrices matrix and what guidelines should be considered when constructing the list of metrics for establishing target specifications in product development? Discuss the importance of each guidelines.	10	L2	3	1
5. a)	Compare and contrast individual brainstorming and group brainstorming.	6	L1	4	1
b)	List and explain the hints for generating solution concepts.	6	L1	4	1
c)	What is the problem decomposition technique? Explain different approaches of problem decomposition.	8	L2	4	1
6. a)	What is benchmarking and why it is required? Illustrate different types of benchmarking.	10	L2	3	1
b)	Explain five-step concept generation method with neat diagram.	10	L2	4	1
Unit – III					
7. a)	Compare the natural life cycle and product life cycle. Illustrate with diagram.	10	L2	5	1
b)	How to apply DFE guidelines to the initial product design? Discuss with example.	10	L2	5	1
8. a)	Identify and explain the internal and external drivers for DFE.	10	L2	5	1
b)	How assess the environmental impacts and compare the environmental impacts to DFE goals.	10	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
